0.1 Challenge 57: LIGO with modified mirrors for ultralight vector dark matter

Problem ID: `Challenge_57_main`
Primary inferred discipline: Gravitation, Cosmology & Astrophysics
Secondary inferred disciplines: High Energy & Nuclear Physics
Python implementation: `challenges/57/answer.py`

Problem

Problem setup

Consider a vector field A^μ coupled to the standard model $B-L$ current J^μ_{B-L} through $\mathcal{L} \supset -\epsilon_{B-L} e J^\mu_{B-L} A_\mu$. If the vector field makes up all the dark matter, the local dark matter density indicates that the field could exert a measurable force on mirrors in laser interferometers, with sufficiently large coupling. Consider LIGO, with a strain sensitivity of $3 \times 10^{-24}\,\mathrm{Hz}^{-1/2}$ at frequency $250\,\mathrm{Hz}$.

Main problem

Take the dark charge Q_D to mass M radio to be $\frac{Q_D}{M} \sim \frac{0.5}{m_n}$ for the inner mirrors, where m_n is the mass of a neutron. We introduce a doped outer mirror, resulting in a differential forced between the inner and outer mirror. We assume the outer mirror gains an additiona $\delta\left(\frac{Q_D}{M}\right) \sim \frac{\delta q}{m_n}$, where m_n is the mass of a neutron. With an observation time of 13 years and SNR of 1, what is the smallest ϵ_{B-L} it can probe at $250\,\mathrm{Hz}$? Provide answers for scenarios where $\delta q = \{0.074, 6 \times 10^{-3}, 5 \times 10^{-4}\}$.

Reviewer's note. The AI-generated solution has been replaced in full by the solution below, which was derived independently before the AI text was read, as the evaluation instructions suggest. The final answer follows the solution. An item-by-item assessment of the challenge statement, of the AI-generated solution and of its numerical script, with the AI text and script reproduced verbatim, follows the final answer in the accompanying document.

Solution

Overview, conventions, and reproducibility

The solution is provided in the following steps:

Step 1: The vector dark-matter field and its dark electric field are obtained from the dark-matter density.
Step 2: The $B-L$ coupling turns the dark electric field into a force on each mirror, and the free-mass response gives its displacement.
Step 3: The exact round trip of the laser light in one arm, with different charge-to-mass ratios for the inner and outer mirror, gives the change of the arm's optical path.
Step 4: The Michelson output is averaged over the unknown polarization and propagation direction of the field.
Step 5: The finite coherence of the field fixes the detection threshold for a 13 yr observation.
Step 6: The smallest detectable ϵ_{B-L} follows for each δq.

Throughout, L is the arm length, $\sqrt{S_h} = 3 \times 10^{-24}\,\mathrm{Hz}^{-1/2}$ the one-sided strain amplitude spectral density at $f = \omega/2\pi = 250\,\mathrm{Hz}$, $T = 13$ yr the observation time, $q_0 = 0.5$ the dark charge-to-mass ratio of the inner mirrors in units of $1/m_n$, $\delta q \in \{0.074,\ 6 \times 10^{-3},\ 5 \times 10^{-4}\}$ the excess of the outer mirrors, m_n the neutron mass, and e the elementary charge. Three conventions are fixed up front. (i) All quantities are evaluated in SI units with ϵ_0 explicit, so that ϵ_{B-L}, which is dimensionless, does not depend on the choice between Gaussian and Heaviside-Lorentz natural units made in the source literature.[1] (ii) The strain sensitivity is a one-sided amplitude spectral density. (iii) 13 yr means 13 Julian years, 4.1025×10^8 s.

Every number below is produced by `challenge57.py`, which lists all problem inputs and all model inputs (with their sources) at its top, uses `astropy` (CODATA 2018) for physical constants, and prints its internal consistency checks under a `-verbose` flag. The problem does not state the dark-matter density, the dark-matter velocity, or the arm length. The values adopted for them are model inputs, chosen and justified in Steps 1, 3, and 5, and their consequences are reported in Table 1.

Step 1: The vector field and its dark electric field

A massive vector field of mass m_A that makes up the dark matter is a superposition of nonrelativistic waves,

$$\mathbf{A}(t, \mathbf{r}) = \sum_i A_i\, \hat{\mathbf{e}}_i \cos(\omega_i t - \mathbf{k}_i \cdot \mathbf{r} + \varphi_i), \qquad \omega_i \simeq \frac{m_A c^2}{\hbar}, \qquad \mathbf{k}_i = \frac{m_A \mathbf{v}_i}{\hbar}, \tag{1}$$

where A_i, $\hat{\mathbf{e}}_i$, $\mathbf{v}_i$ and φ_i are the amplitude, polarization unit vector, velocity and phase of the i-th wave [1, 2]. At $f = 250\,\mathrm{Hz}$, $m_A c^2 = hf = 1.0339 \times 10^{-12}\,\mathrm{eV}$. Over one coherence time (Step 5) the sum acts as a single wave of amplitude A_0 with $\rho = \frac{1}{2} m_A^2 A_0^2$ in natural units, and the dark electric field $\mathbf{E}_D = -\partial_t \mathbf{A}$ has amplitude $m_A A_0 = \sqrt{2\rho}$, independent of m_A. In SI units

$$E_D = \sqrt{\frac{2\rho}{\epsilon_0}}. \tag{2}$$

[1] Pierce et al. [1] use $\hbar = c = \sqrt{4\pi\epsilon_0} = 1$ and Morisaki et al. [2] use $\hbar = c = \epsilon_0 = 1$. The force $\epsilon e Q_D E_D$ is the same physical quantity in both once the field amplitude is expressed through the energy density in the same units.

The problem prompt does not explicitly provide ρ; we instead adopt $\rho = 0.4\,\mathrm{GeV\,cm^{-3}}$ as given in Challenge 56, which is also the value commonly quoted in interferometer dark-photon analyses [1–3]. This choice gives $E_D = 3804.8\,\mathrm{V\,m^{-1}}$.[2]

Step 2: Force on a mirror and its displacement

The interaction $\mathcal{L} \supset -\epsilon_{B-L}\, e\, J^{\mu}_{B-L} A_\mu$ produces, in the nonrelativistic limit, a force $\mathbf{F} = \epsilon_{B-L} e Q_D \mathbf{E}_D$ on a body of $B-L$ charge Q_D, while the magnetic counterpart is suppressed by v/c [1]. For a neutral atom the $B-L$ charge is its neutron number, so $Q_D/M = (A-Z)/(\mu\, m_n)$ with A, Z and μ the mass number, atomic number and atomic mass in atomic units. This is $0.501/m_n$ for fused silica and $0.51/m_n$ for sapphire [7], as follows from the standard atomic weights [8], and the problem's $q_0/m_n = 0.5/m_n$ for the inner mirrors is the value for the fused-silica test masses of Advanced LIGO [9]. At 250 Hz, far above the pendulum resonance of its suspension, a mirror responds to the force as a free mass, so its displacement along the polarization is

$$\delta x(t) = \frac{a}{\omega^2} \sin(\omega t - \mathbf{k}\cdot\mathbf{r} + \varphi), \qquad a = \epsilon_{B-L}\, e\, \frac{Q_D}{M}\, E_D. \tag{3}$$

Per unit ϵ_{B-L} the inner mirrors have $e\, q_0/m_n = 4.7828 \times 10^7\,\mathrm{C\,kg^{-1}}$, $a_0 = 1.8197 \times 10^{11}\,\mathrm{m\,s^{-2}}$ and $\delta x_0 = a_0/\omega^2 = 7.3751 \times 10^4\,\mathrm{m}$. The outer mirrors have these multiplied by $(q_0 + \delta q)/q_0$.

Step 3: Exact round trip in one arm with a doped outer mirror

Let the inner (input) and outer (end) mirrors of an arm along the unit vector $\hat{\mathbf{n}}$ sit at $x_i(t) = \hat{\mathbf{n}} \cdot \delta\mathbf{x}_i(t)$ and $x_e(t) = L + \hat{\mathbf{n}} \cdot \delta\mathbf{x}_e(t)$. Light leaving the inner mirror at $t - 2L/c$ reflects from the outer mirror at $t - L/c$ and returns at t, so the round-trip path is [2, Eq. 11], [7, Eq. 9]

$$c\, T_r(t) = -x_i(t) + 2x_e(t - L/c) - x_i(t - 2L/c). \tag{4}$$

With $\delta\mathbf{x}_e = (1 + \delta q/q_0)\,\delta\mathbf{x}$ evaluated at the outer mirror and $\delta\mathbf{x}_i = \delta\mathbf{x}$ at the inner mirror, and $\delta\mathbf{x}(t) \propto \sin\omega t$, the second difference of the common part gives $-\sin\omega t + 2\sin\omega(t - L/c) - \sin\omega(t - 2L/c) = 4\sin^2(\omega L/2c)\,\sin\omega(t - L/c)$, so the change of the round-trip path is

$$\delta L(t) = \Big[\, \underbrace{4\sin^2\!\Big(\frac{\omega L}{2c}\Big)}_{\text{light-travel time}} + \underbrace{\frac{2\,\delta q}{q_0}}_{\text{doping}} \,\Big]\, \hat{\mathbf{n}} \cdot \delta\mathbf{x}(t - L/c) \; + \; \underbrace{2L\,(\hat{\mathbf{n}}\cdot\mathbf{k})\,(\hat{\mathbf{n}}\cdot\delta\mathbf{x}_0)\cos\omega(t - L/c)}_{\text{field gradient}}. \tag{5}$$

The light-travel-time term is the default signal, present without any doping. Because the light samples the two co-moving mirrors at times $2L/c$ apart, the optical path changes even though their separation does not [2]. It is the second finite difference in time of the common displacement, $-x(t) + 2x(t - L/c) - x(t - 2L/c)$, which, for the sinusoidal mirror motion $\delta x \propto \sin\omega t$ driven by the field, equals $4\sin^2(\omega L/2c)\,\delta x(t - L/c) \approx (\omega L/c)^2\,\delta x$ and is therefore second order in $\omega L/c$. The doping term is the differential force introduced by the doping. The two terms oscillate in phase and therefore add constructively. Lastly, the field-gradient term is the phase difference of the field across the arm, the original signal of Pierce et al. [1], in quadrature with the others and suppressed by $kL = m_A v L/\hbar$. With $L = 3994.5\,\mathrm{m}$ [9],[3] $\omega L/c = 2.0930 \times 10^{-2}$, $4\sin^2(\omega L/2c) = 4.380 \times 10^{-4}$, $2\delta q/q_0 = 0.296,\ 0.024,\ 0.002$, and $2kL = 3.21 \times 10^{-5}$ for $v = 230\,\mathrm{km\,s^{-1}}$ (Step 5). The undoped light-travel-time term is thus 0.15%, 1.8% and 22% of the doping term for the three values of δq: never dominant, but not negligible at the precision requested, and for the smallest δq it changes the answer by a fifth (Fig. 1).

[2] The direct-detection convention is $\rho = 0.3\,\mathrm{GeV\,cm^{-3}}$ [4, 5]. Since $\epsilon_{B-L,\min} \propto \rho^{-1/2}$, that choice raises every answer by 15% (Table 1). The amplitude in any one coherence patch is Rayleigh distributed about the value implied by ρ [6]. Over the $\sim 6 \times 10^4$ coherence times of a 13 yr observation this averages out.

[3] The problem says only "LIGO". Using the nominal 4000 m instead changes the answers by 0.13% (Table 1).

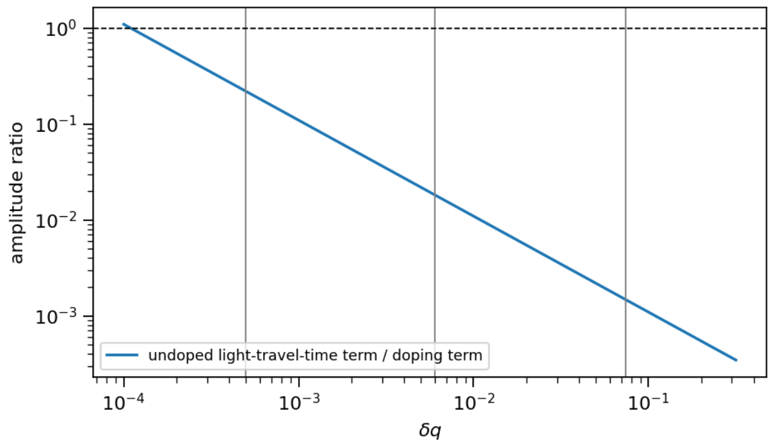

Figure 1: Ratio of the undoped light-travel-time term to the doping term in Eq. (5) as a function of δq at 250 Hz and $L = 3994.5$ m. Vertical lines mark the three values of δq in the problem. The dashed line marks equality, reached only at $\delta q = 1.1 \times 10^{-4}$.

Step 4: Michelson output averaged over the field orientation

The detector output is $h(t) = -[\delta L_{\hat{\mathbf{n}}}(t) - \delta L_{\hat{\mathbf{m}}}(t)]/(2L)$ for arms along $\hat{\mathbf{n}}$ and $\hat{\mathbf{m}}$ with $\hat{\mathbf{n}} \cdot \hat{\mathbf{m}} = 0$ [2, Eq. 9]. The Fabry-Perot arm cavities amplify any change of the arm length, and that amplification is already contained in the strain sensitivity [2]. The polarization and propagation direction of the field are unknown and, over 13 yr, are sampled through the rotation of the Earth, so the relevant quantity is the mean square over an isotropic distribution of $\hat{\mathbf{e}}$ and $\hat{\mathbf{k}}$, with $\langle(\hat{\mathbf{n}} \cdot \hat{\mathbf{e}})^2\rangle = 1/3$ and no correlation between $\hat{\mathbf{e}}$ and $\hat{\mathbf{k}}$ [1, 2, 7].[4] Writing $\Gamma \equiv 4\sin^2(\omega L/2c) + 2\delta q/q_0$ and averaging Eq. (5) over time and orientation,

$$\langle h^2 \rangle = \frac{\delta x_0^2}{4L^2}\left[\frac{\Gamma^2}{3} + \frac{(2kL)^2}{9}\right]\epsilon_{B-L}^2. \tag{6}$$

For $\delta q = 0$ this reproduces Eqs. (23) and (24) of Morisaki et al. [2] exactly, fixing the averaging factors independently of our derivation. Per unit ϵ_{B-L}, $\langle h^2\rangle^{1/2} = 1.5800$, 0.13025 and 0.012995 for the three values of δq. The field-gradient term has a relative contribution $\leq 10^{-4}$ in all three cases.

Step 5: Detection threshold for a 13 yr observation

The field is coherent only over $\tau = 2\pi/(m_A v^2)$, the time over which waves differing by the local dark-matter velocity dispersion v (the notation of the source literature) dephase [2, Eq. 5]. The problem does not give v. We adopt $v = 230\,\mathrm{km\,s^{-1}}$ as provided in the Challenge 56 prompt, which is also the value commonly adopted in the literature [1, 2, 7]. This gives $\tau = 6796$ s, so the 13 yr observation spans $T/\tau = 6.0 \times 10^4$ coherence times.[5] Beyond one coherence time the phase of the field becomes stochastic, so the signal cannot

[4] Two distinct averages are involved, over the propagation direction $\hat{\mathbf{k}}$ and over the polarization $\hat{\mathbf{e}}$. Neither is isotropic by kinematics alone: the halo wind is ordered and the velocity ellipsoid of a virialized halo is anisotropic, and the Earth's rotation does not randomize a fixed $\hat{\mathbf{e}}$. The average over $\hat{\mathbf{k}}$ is nevertheless irrelevant at the precision requested, since $\hat{\mathbf{k}}$ enters only the field-gradient term, below 10^{-4} of the signal here. The average over $\hat{\mathbf{e}}$ is a genuine assumption about the field, that its polarization is randomized between the $\sim 6 \times 10^4$ coherence patches sampled in 13 yr. It is the convention under which all published limits are quoted [1, 2, 10], and Michimura et al. [7] note that a production mechanism leaving only longitudinal or only transverse modes would violate it. Its exposure is $\mathcal{O}(1)$: the in-phase amplitude scales as $|(\hat{\mathbf{n}} - \hat{\mathbf{m}}) \cdot \hat{\mathbf{e}}|$, whose isotropic root mean square $\sqrt{2/3}$ would become $\sqrt{2}$ for a fixed polarization along $\hat{\mathbf{n}} - \hat{\mathbf{m}}$ and 0 for one normal to the detector plane.

[5] As in any wave-like dark-matter search, the line shape depends on two velocities, the dispersion of the halo and the streaming speed of the detector through it, and $230\,\mathrm{km\,s^{-1}}$ is the galactic rotation speed of the classic halo model [4]. The source papers use it in both roles, as do we.

Table 1: Smallest detectable ϵ_{B-L} at 250 Hz, $T = 13$ yr, for the three values of δq. "Exact" is Eq. (5) with all three terms, $L = 3994.5$ m, $\rho = 0.4\,\mathrm{GeV\,cm^{-3}}$, $v = 230\,\mathrm{km\,s^{-1}}$, and the power threshold of Eq. (7).

signal model	threshold	$\delta q = 0.074$	6×10^{-3}	5×10^{-4}
exact (this work)	Eq. (7)	1.47×10^{-27}	1.78×10^{-26}	1.79×10^{-25}
doping term only (as the problem implies)	Eq. (7)	1.47×10^{-27}	1.82×10^{-26}	2.18×10^{-25}
exact	amplitude matched filter, $R(T)$	1.34×10^{-27}	1.63×10^{-26}	1.63×10^{-25}
exact, $L = 4000$ m	Eq. (7)	1.47×10^{-27}	1.78×10^{-26}	1.79×10^{-25}
exact, $\rho = 0.3\,\mathrm{GeV\,cm^{-3}}$	Eq. (7)	1.70×10^{-27}	2.06×10^{-26}	2.06×10^{-25}

be integrated coherently over the full T. The appropriate search instead sums the power of successive Fourier transforms of length $\sim\tau$ [10, 11]. The sensitivity still improves with observation time, but only as $T^{1/4}$ in amplitude rather than the $T^{1/2}$ of a coherent signal, because the powers of the $N = T/\tau$ independent stretches add with $\sqrt{N}$ fluctuations. The resulting detection threshold is [2, Eqs. 20 and 21], [1, Eq. 9]

$$\langle h^2\rangle_{\rm thr} = \frac{S_h(f)}{T_{\rm eff}}, \qquad T_{\rm eff} = \sqrt{\tau T} = 1.6697\times10^{6}\,\mathrm{s}, \qquad \langle h^2\rangle_{\rm thr}^{1/2} = 2.3217\times10^{-27}, \tag{7}$$

which is the statement that the signal power equals the noise power in the power statistic, i.e., a power signal-to-noise ratio of one. We adopt this as the meaning of "SNR = 1" because it is the convention of the literature the problem is built on.[6]

Step 6: The coupling detection threshold

Combining Eqs. (6) and (7),

$$\epsilon_{B-L,\rm min} = \left[\frac{S_h/T_{\rm eff}}{\langle h^2\rangle/\epsilon_{B-L}^2}\right]^{1/2} = 1.47\times10^{-27}, \quad 1.78\times10^{-26}, \quad 1.79\times10^{-25} \tag{8}$$

for $\delta q = 0.074$, 6×10^{-3} and 5×10^{-4}. Table 1 shows how the forecast thresholds vary under the alternative assumptions discussed above. For scale, the same formula with no doping at all gives $\epsilon_{B-L,\rm min}\approx 1\times10^{-24}$ for 13 yr, the undoped reach of Morisaki et al. [2] extended in time. The doping improves it by $2\delta q/q_0$ divided by 4.4×10^{-4}.

What the problem idealizes

(i) *The undoped signal.* The problem attributes the differential force to the doping alone. The finite light-travel time produces a differential signal even for identical mirrors, in phase with the doping term. It is included in the primary answer and is what separates the first two rows of Table 1. (ii) *Realizability of δq.* The $B-L$ charge-to-mass ratios of the usual optics differ by about 0.01 (silica 0.501, sapphire 0.51; Michimura et al. 7), so $\delta q = 6\times10^{-3}$ is a silica-versus-sapphire-scale difference and 5×10^{-4} an isotopic-doping level, whereas $\delta q = 0.074$ would require a mirror substrate of a heavy element such as indium, cadmium or tin, whose neutron fractions $(A-Z)/\mu$ computed from the standard atomic weights [8] are 0.57 to 0.58, not a dopant. (iii) *Provenance.* To our knowledge the configuration has not been proposed in the published literature. The closest precedents are the composition dipoles of equivalence-principle tests, mirrors of different thickness proposed for scalar dark matter [13], the different response of KAGRA's

[6] It is not the only possible reading. In Challenge 56 we defined SNR through the amplitude matched filter, $h_0\sqrt{T/S_h} = 1$ for a monochromatic signal, and carried the finite coherence through the exact factor $R(T)$ derived from the field correlation function of Derevianko [12]. With $\langle h^2\rangle = h_0^2/2$ that convention gives $R(T) = 20.28$ here and thresholds 8.5% lower (Table 1). The two differ by $\mathcal{O}(1)$ factors inherent to defining an SNR for a stochastic signal, and the problem statement does not choose between them.

sapphire test masses and fused-silica auxiliary mirrors exploited through auxiliary length channels [7], and the use of different materials in an optomechanical accelerometer [14]. It is a natural extension of these, and nothing in the calculation depends on its provenance; we just flag this for assessment completeness. (iv) *Thirteen years of continuous data.* The threshold of Eq. (7) assumes 13 yr of uninterrupted data at a constant sensitivity, whereas real detectors have duty cycles (the fraction of calendar time spent taking science data) well below 100% and are upgraded between observing runs, so that the sensitivity improves over such a span. It also treats v as constant, although the Earth's orbital motion modulates the lab-frame speed by about $\pm 15\,\mathrm{km\,s^{-1}}$ over the year, the component of the $30\,\mathrm{km\,s^{-1}}$ orbital velocity along the halo wind [4, Eq. 3.6], and hence τ by about $\pm 13\%$, and it uses the mean-square field amplitude, although the amplitude fluctuates from one coherence patch to the next. Both effects average over the 6×10^4 coherence patches of the observation.

Is the problem well posed?

As a statement it is: the requested quantity is defined for each δq. It is not closed in its inputs: the dark-matter density, the dark-matter velocity (which sets the coherence time and hence $\epsilon_{\min} \propto v^{1/2}$) and the arm length are not given and must be supplied externally, here inherited from the Challenge 56 prompt (ρ, v) or taken from the instrument literature (L), and the mirror material is only implied by $q_0 = 0.5$. Nor is it closed in its conventions: "SNR = 1" for a stochastic, incoherently summed signal is a power statistic in the problem's source literature but an amplitude statistic in others, an $\mathcal{O}(1)$ difference (8.5% here). The statement also omits one piece of physics, the undoped light-travel-time signal of Step 3. It raises the signal by 0.15% and 1.8% for the two larger δq but by 22% for the smallest, for which the answer is lowered by 18%. At three significant digits the answers therefore depend on these choices. The exact chain above states each one and Table 1 quantifies each. Agreement with a reference answer beyond the second digit would test which conventions the reference adopted, not the physics.

Filled `template.py`

Listing 1: `template.py`

```python
def answer():
    r"""
    Return the values of the smallest $\epsilon_{B-L}$ for three scenarios.

    Inputs
    ----------
    None

    Outputs
    ----------
    eps_B_L_min: list[float],
          smallest $\epsilon_{B-L}$ it can probe at $250\,\text{Hz}$ with an observation time
    of $13$ years and SNR of $1$
          for scenarios where $\delta q=\{0.074, 6\times 10^{-3}, 5\times 10^{-4}\}$.
    """

    # ------------------ FILL IN YOUR RESULTS BELOW ------------------
    # Dimensionless.  Computed by Claude_Notes/challenge57.py (exact round-trip signal with
    doping,
    # finite light-travel time and field gradient; rho = 0.4 GeV/cm^3, v = 230 km/s, L = 3994.5
     m;
    # power threshold <h^2> = S_h / sqrt(tau T)).  Doping term only: [1.47e-27, 1.82e-26, 2.18e
    -25].
    eps_B_L_min = [1.47e-27, 1.78e-26, 1.79e-25]
```

```
    # ------------------------------------------------------------

    return eps_B_L_min
```

References

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Final answer

$$\boxed{\epsilon_{B-L,\mathrm{min}} = 1.47 \times 10^{-27}, \qquad 1.78 \times 10^{-26}, \qquad 1.79 \times 10^{-25}}$$

Assessment of the challenge statement and of the AI-generated solution

Scope, provenance, and method

The files assessed are those supplied for Challenge 57: `problem.tex`, the AI-generated `solution.tex`, `final_answer.tex`, `numerics.py`, `answer.py`, `metadata.json` and `agent_log.md`. None of them was modified. The AI solution and its script are reproduced verbatim in Appendices A and B. The solution was originally generated by `codex-cli/gpt-5.6-sol` at reasoning effort "high" on 2026-08-15 and is marked "generated-unreviewed". The agent had web access through a server-side search tool (11 searches, queries logged, returned pages not), including a direct fetch of Michimura et al. [7] (`arxiv.org/html/2008.02482v3`), and a Python shell without network access (2 commands, no scratch file). Its `numerics.py` runs (Python 3.11) and reproduces its boxed values 5.4278×10^{-29}, 6.6943×10^{-28} and 8.0332×10^{-27} to twelve digits.

The AI solution is assessed against the reference solution above. The AI-generated results, including the intermediate numbers in its agent log, are all reproduced exactly by adopting the AI's own model in an independent script, so that every discrepancy below is attributed to a named modeling choice and not to arithmetic.

The challenge statement

The statement specifies the coupling, the mirror charge-to-mass ratios, the frequency, the noise level, the observation time and the SNR, and asks for three numbers. It does not specify five inputs that the answer needs.

(i) The local dark-matter density. Both solutions take $0.4\,\mathrm{GeV\,cm^{-3}}$, the value of the source literature [1–3] and of the companion Challenge 56. The alternative $0.3\,\mathrm{GeV\,cm^{-3}}$ [5] raises the answer by 15.5%.

(ii) The dark-matter velocity, which sets the coherence time $\tau = 2\pi/(m_A v^2)$ and thereby the threshold ($\epsilon_{\min} \propto v^{1/2}$). The reference takes $230\,\mathrm{km\,s^{-1}}$; the AI agent used $300\,\mathrm{km\,s^{-1}}$ in its log and no value in its text. The difference is 14%.

(iii) The arm length (3994.5 m [9]; the AI took 4000 m, 0.13%).

(iv) The mirror material, needed only for the signal of the undoped baseline through the finite light-travel time (Step 3), which is 18% of the answer at $\delta q = 5 \times 10^{-4}$ and negligible at $\delta q = 0.074$.

(v) The detection statistic. "13 years" with "SNR = 1" does not say how a 13 yr record is combined when the field is coherent for only 6800 s. The source literature uses the semicoherent power threshold $\langle h^2 \rangle = S_h/\sqrt{\tau T}$ [1, 2], adopted in Step 5; the amplitude statistic with the finite-coherence factor of Challenge 56 gives 8.5% less. A fully coherent integration is not a convention but an error (see below).

The doped-mirror configuration itself has no published precedent (the AI agent searched for the literal values of δq and found none, consistent with our independent searches), but it is a natural extension of the material-asymmetry proposals of Michimura et al. [7] and Manley et al. [14] and is well defined. "13 years" is an idealization of continuous data at fixed sensitivity, which no observing program delivers, but it is a stated input and is used as such. The statement therefore determines the answer to about 20% once the five inputs are fixed at their literature values, and not to the three figures a boxed answer suggests. It cannot be corrected unambiguously, so `problem.tex` is left untouched and the assumptions are stated in the solution.

The AI solution, item by item

Table 2 lists every modeling element of the AI solution against the reference solution, with the same two classification axes as for Challenge 56, recorded in the "Classification" column: hidden curriculum versus solver (AI) error versus prompt deficiency, and consequential versus minor versus simplification. The "Effect" column gives the factor by which that item alone changes the AI answer, on which the severity is judged.

Table 2: The AI solution for Challenge 57 against the reference solution. "Effect" is the factor by which the item alone lowers the AI's threshold $\epsilon_{B-L,\mathrm{min}}$ relative to the reference solution (a factor below one means the AI claims a better sensitivity).

Item	AI solution	Reference	Classification	Effect
Dark electric field $E_D = \sqrt{2\rho}$, independent of m_A; $m_A c^2 = hf$	correct, natural units	same, Step 1	correct	none
Differential charge-to-mass ratio $\delta q/m_n$; base $0.5/m_n$ cancels	correct	same, Step 2	correct within a doping-only model	none
Free-mass displacement $\Delta a_0/m_A^2$	correct	same	correct	none
Angular projection	"optimal projection of the vector force along the doped arm", one arm, $h_0 = \Delta x_0/L$	isotropic average over the field polarization and both arms, $\langle h^2\rangle^{1/2} = \Gamma\delta x_0/(2\sqrt{3}L)$, Step 4	simplification, stated, but unphysical for a 13 yr search: the Earth's rotation sweeps the arm relative to any fixed polarization, and the second arm's doped mirror enters the output with opposite sign. The source literature averages [1, 2]. Consequential.	0.577 ($1/\sqrt{3}$)
Finite light-travel-time signal of the undoped baseline, $4\sin^2(\omega L/2c)$	omitted, not mentioned	included exactly, Step 3, with the field-gradient term	hidden curriculum bordering on omission: the effect is in Morisaki et al. [2] and in Eq. (9) of Michimura et al. [7], which the agent fetched. Simplification, unstated.	1.00, 1.02, 1.22
Detection threshold	fully coherent, $h_{0,\mathrm{min}} = \sqrt{S_h/T}$ with $T = 13$ yr, stated as an assumption	semicoherent, $\langle h^2\rangle_{\mathrm{thr}} = S_h/\sqrt{\tau T}$, $\tau = 6796$ s, $T_{\mathrm{eff}} = 1.67 \times 10^6$ s, Step 5	genuine error, consequential. Not hidden curriculum: the coherence limit is in Pierce et al. [1], Eq. (9), and in Guo et al. [3], which the AI cites for its conventions. The AI's own last paragraph calls the coherent assumption "idealized" and says the real reach is "weaker", and its agent log had adopted the semicoherent threshold (see below).	0.064 (1/15.7)
Density $\rho = 0.4\,\mathrm{GeV\,cm^{-3}}$	assumed and stated	same	prompt deficiency, handled correctly	none

continued on next page

Table 2, continued

Item	AI solution	Reference	Classification	Effect
Arm length	4000 m, cited to the LIGO Laboratory web page	3994.5 m [9]	simplification, stated; minor	1.001
Velocity	absent from the text; $300\,\mathrm{km\,s^{-1}}$ in the log	$230\,\mathrm{km\,s^{-1}}$	prompt deficiency; no effect on the AI's coherent answer	none
Constants	m_n and α are CODATA 2022 values while the citation is the NIST CODATA compilation	CODATA 2018	minor, 10^{-9}	none
Precision	twelve digits in the box	three, with the convention band stated	minor, presentation	none
Internal consistency	the assumption list says "fully coherent"; the closing paragraph disowns it	consistent	consequential for usability: the boxed number is one the text itself declares unrealistic	
Net	5.43×10^{-29}, 6.69×10^{-28}, 8.03×10^{-27}	1.47×10^{-27}, 1.78×10^{-26}, 1.79×10^{-25}		1/27.1, 1/26.6, 1/22.2

The AI values overstate the reach by factors of 27, 27 and 22. Adopting the AI's assumptions in an independent script (SciPy CODATA constants, natural units, $L = 4000$ m, $\rho = 0.4\,\mathrm{GeV\,cm^{-3}}$, coherent threshold, $h_0 = \Delta x_0/L$) reproduces all its output values to better than a relative 4×10^{-10} (the residual being the CODATA 2018 versus 2022 constants). The full discrepancy is thus accounted for by three identifiable choices: the coherent integration (15.7), the optimal projection ($\sqrt{3}$), and, at the smallest δq, the dropped light-travel-time term (1.22). The arithmetic and the unit chain of the AI solution are correct. Its references are real and relevant [3].

The AI numerical script

`numerics.py` (Appendix B) is under forty lines and mirrors the text with exact constants, and it asserts agreement with `answer.py` to 5×10^{-7}. It runs and passes. It contains no coherence time, no averaging, no light-travel-time term and no variant: it encodes the same idealization as the text and cannot reveal it. There is no inputs block and no sanity check. As numerical work it is correct for its model and minimal.

Evidence from the agent log

The log shows that the error occurred between the computation and the write-up, not for lack of knowledge. Its central note reads: "The key long-duration point is that the 250 Hz field is not phase-coherent for 13 years: its coherence time is only about 4.0×10^3 s. I'm therefore using the standard semicoherent sensitivity, $T_{\mathrm{eff}} = \sqrt{T_{\mathrm{obs}}\tau}$, and will state the geometric convention explicitly." Its first shell command then computed, with $v = 10^{-3}c$ ($\tau = 4000$ s), $T_{\mathrm{eff}} = \sqrt{\tau T} = 1.281 \times 10^6$ s, a threshold $2\sqrt{S_h}/\sqrt{T_{\mathrm{eff}}} = 5.30 \times 10^{-27}$, and three sets of thresholds: "optimal" 1.94×10^{-27}, 2.39×10^{-26}, 2.87×10^{-25}; "one-arm averaged" ($1/\sqrt{3}$) 3.36×10^{-27}, 4.14×10^{-26}, 4.97×10^{-25}; and "two-arm averaged" ($\sqrt{2/3}$) 2.38×10^{-27}, 2.93×10^{-26}, 3.52×10^{-25}. The same command also printed the "coherent naive values" 1.08×10^{-28}, 1.34×10^{-27}, 1.60×10^{-26} for comparison. Its second command repeated the semicoherent values for $L = 4000$ m (3.36×10^{-27}, 4.15×10^{-26}, 4.98×10^{-25} one-arm averaged; 1.94×10^{-27}, 2.40×10^{-26}, 2.88×10^{-25} optimal). All of these are reproduced exactly by our independent script under these stated assumptions. The reasoning headers recorded after these computations (the short titles the agent emits for each reasoning step) are "Deciding angular-averaged one-arm presentation", "Aligning citations and refining constants" and "Calculating differential acceleration from mirror doping": the agent was preparing to present the angular-averaged semicoherent result, and no header records a decision to revert to the coherent idealization. The delivered text nevertheless boxes the fully coherent, optimally projected values, which appear nowhere in the log, with a threshold $\sqrt{S_h/T}$ that differs from the log's "coherent naive" convention by a further factor of two. Had the agent delivered its own one-arm-averaged semicoherent numbers, they would have agreed with the reference solution to within a factor of 2.3 to 2.8, the remaining differences being the velocity (300 versus $230\,\mathrm{km\,s^{-1}}$), the factor-two threshold convention, the single-arm projection and the light-travel-time term. This record is useful for the project's downstream analysis: the failure mode is a late substitution of an idealization the solver had already rejected, with the rejection surviving only as a caveat in the last sentence.

Verdict

This AI-generated attempt is objectively poor as a reference solution and adequate only as an internally consistent idealized estimate. The field-to-force-to-displacement chain and the units are right. The two decisive physics inputs, the treatment of the coherence time and the angular projection, are wrong or unphysical, and the text knows it for the first. The script is correct for its model and minimal, and it shares the model's blindness. Holistically the boxed values are wrong by a factor of 22 to 27, from a genuine error of judgment (a knowing choice of the coherent idealization) compounded by an unphysical stated assumption; the remaining omissions (light-travel-time term, arm length) are minor by comparison.

Recommended reference values and tolerance for grading

Reference: $\epsilon_{B-L,\min} = 1.47\times10^{-27}$, 1.78×10^{-26}, 1.79×10^{-25} for $\delta q = 0.074$, 6×10^{-3}, 5×10^{-4} (exact round-trip signal, isotropic average, semicoherent power threshold, $\rho = 0.4\,\mathrm{GeV\,cm^{-3}}$, $v = 230\,\mathrm{km\,s^{-1}}$, $L = 3994.5$ m). Accepted alternatives: the doping-only values 1.47×10^{-27}, 1.82×10^{-26}, 2.18×10^{-25} (the reading the statement suggests), and shifts of -8.5% (amplitude statistic with finite-coherence factor), $+15.5\%$ ($\rho = 0.3\,\mathrm{GeV\,cm^{-3}}$) and -7% to $+14\%$ (velocity between 200 and $300\,\mathrm{km\,s^{-1}}$). A tolerance of $\pm25\%$ around the reference solution, or around the doping-only values for the smallest δq, therefore covers every defensible convention. A value smaller by a factor of about 1.7 indicates the optimal-projection assumption alone (partially correct). Values smaller by a factor of ten or more indicate a fully coherent 13 yr integration and are wrong.

Appendix A: the AI-generated solution and final answer, verbatim

Solution

Step-by-Step Derivation

Work in natural units, $\hbar = c = 1$, and assume:

- canonical local density $\rho_{\rm DM} = 0.4\,\mathrm{GeV\,cm^{-3}}$;
- LIGO arm length $L = 4\,\mathrm{km}$;
- optimal projection of the vector force along the doped arm;
- fully coherent matched-filter integration for $T = 13$ years.

The density and force conventions follow the LIGO dark-photon analysis, which gives $A_0 = \sqrt{2\rho_{\rm DM}}/m_A$ and $a = \epsilon e(Q_D/M)\dot{A}$ Guo et al.. LIGO's arms are each 4 km long LIGO Laboratory.

1. Dark electric-field amplitude

Write the nonrelativistic vector field as

$$\mathbf{A}(t) = \mathbf{A}_0 \cos(m_A t).$$

Its energy density is

$$\rho_{\rm DM} = \frac{1}{2} m_A^2 A_0^2,$$

so

$$A_0 = \frac{\sqrt{2\rho_{\rm DM}}}{m_A}.$$

The oscillating dark electric field has amplitude

$$E_{D,0} = |\dot{\mathbf{A}}|_0 = m_A A_0 = \sqrt{2\rho_{\rm DM}}.$$

2. Differential acceleration

The inner and doped outer mirrors have

$$\left(\frac{Q_D}{M}\right)_{\rm inner} = \frac{0.5}{m_n},$$

$$\left(\frac{Q_D}{M}\right)_{\rm outer} = \frac{0.5+\delta q}{m_n}.$$

Therefore the common $0.5/m_n$ contribution cancels, leaving

$$\Delta\left(\frac{Q_D}{M}\right) = \frac{\delta q}{m_n}.$$

Thus the differential acceleration amplitude is

$$\Delta a_0 = \epsilon_{B-L} e \,\frac{\delta q}{m_n} \sqrt{2\rho_{\rm DM}}.$$

3. Displacement and strain

For a freely suspended mirror driven at angular frequency $\omega = m_A$,

$$\Delta x_0 = \frac{\Delta a_0}{m_A^2}.$$

The resulting strain is

$$h_0 = \frac{\Delta x_0}{L} = \epsilon_{B-L}\delta q\,\frac{e\sqrt{2\rho_{\rm DM}}}{m_n m_A^2 L}.$$

At $f = 250\,\mathrm{Hz}$,

$$m_A = hf = (4.135667696 \times 10^{-24}\,\mathrm{GeV\,s})(250\,\mathrm{s}^{-1}) = 1.033916924 \times 10^{-21}\,\mathrm{GeV}.$$

Using

$$1\,\mathrm{cm} = 5.06773071616 \times 10^{13}\,\mathrm{GeV}^{-1},$$

we obtain

$$\rho_{\rm DM} = \frac{0.4}{(5.06773071616 \times 10^{13})^3} = 3.07340223055 \times 10^{-42}\,\mathrm{GeV}^4.$$

Also,

$$m_n = 0.93956542194\,\mathrm{GeV}, \qquad e = \sqrt{4\pi\alpha} = 0.302822120769,$$

with the neutron mass and fundamental constants taken from NIST CODATA, and

$$L = 4000\,\mathrm{m} = 2.02709228646 \times 10^{19}\,\mathrm{GeV}^{-1}.$$

Consequently,

$$\frac{e\sqrt{2\rho_{\rm DM}}}{m_n m_A^2 L} = 36.8757159102,$$

so the signal strain is

$$h_0 = 36.8757159102\,\epsilon_{B-L}\delta q.$$

4. Coherent strain threshold

For a monochromatic signal and one-sided strain noise amplitude spectral density $S_h^{1/2}$,

$$\mathrm{SNR}^2 = \frac{h_0^2 T}{S_h}.$$

At $\mathrm{SNR} = 1$,

$$h_{0,\rm min} = \frac{S_h^{1/2}}{\sqrt{T}}.$$

The observation time is

$$T = 13(365.25)(86400) = 410248800\,\mathrm{s}.$$

Therefore,

$$h_{0,\mathrm{min}} = \frac{3 \times 10^{-24}}{\sqrt{410248800}} = 1.48114506204 \times 10^{-28}.$$

Solving for the minimum coupling,

$$\epsilon_{B-L,\mathrm{min}} = \frac{h_{0,\mathrm{min}}}{36.8757159102\,\delta q}.$$

This gives

δq	$\epsilon_{B-L,\mathrm{min}}$
0.074	$5.42781972291 \times 10^{-29}$
6×10^{-3}	$6.69431099159 \times 10^{-28}$
5×10^{-4}	$8.03317318991 \times 10^{-27}$

The fully coherent 13-year assumption is idealized: virialized vector dark matter near 250 Hz has a coherence time of only approximately 10^3-10^4 s, as discussed in the LIGO analysis. A realistic phase-incoherent search would therefore have a weaker, analysis-dependent reach.

Final Answer

$$\boxed{\epsilon_{B-L,\mathrm{min}} = \left(5.42781972291 \times 10^{-29}, 6.69431099159 \times 10^{-28}, 8.03317318991 \times 10^{-27}\right)}$$

Appendix B: the AI-generated numerical script, verbatim

Listing 2: `numerics.py` (AI-generated)

```
#!/usr/bin/env python3
"""Compute the idealized coherent LIGO dark-vector reach in Challenge 57."""

from __future__ import annotations

import math
import runpy
from pathlib import Path


def sensitivities() -> list[float]:
    cm_in_inverse_gev = 5.06773071616e13
    meter_in_inverse_gev = 5.06773071616e15
    density = 0.4 / cm_in_inverse_gev**3
    mass = 4.135667696e-24 * 250.0
    neutron_mass = 0.93956542194
    alpha = 7.2973525643e-3
    electric_charge = math.sqrt(4.0 * math.pi * alpha)
    arm_length = 4000.0 * meter_in_inverse_gev
    strain_coefficient = (
        electric_charge * math.sqrt(2.0 * density) / (neutron_mass * mass**2 * arm_length)
    )

    observation_time = 13.0 * 365.25 * 86400.0
    strain_threshold = 3.0e-24 / math.sqrt(observation_time)
    return [strain_threshold / (strain_coefficient * delta_q) for delta_q in (0.074, 0.006,
    0.0005)]


def main() -> None:
    values = sensitivities()
    for delta_q, value in zip((0.074, 0.006, 0.0005), values):
        print(f"delta_q={delta_q:g}: epsilon={value:.12e}")
    expected = runpy.run_path(Path(__file__).with_name("answer.py"))["answer"]()
    if any(not math.isclose(value, target, rel_tol=5.0e-7) for value, target in zip(values,
    expected)):
        raise AssertionError((values, expected))


if __name__ == "__main__":
    main()
```

Listing 3: `answer.py` (AI-generated)

```
def answer():
    r"""
    Return the values of the smallest $\epsilon_{B-L}$ for three scenarios.

    Inputs
    ----------
    None
```

```
    Outputs
    ----------
    eps_B_L_min: list[float],
          smallest $\epsilon_{B-L}$ it can probe at $250\,\text{Hz}$ with an observation time
    of $13$ years and SNR of $1$
          for scenarios where $\delta q=\{0.074, 6\times 10^{-3}, 5\times 10^{-4}\}$.
    """

    # ------------------ FILL IN YOUR RESULTS BELOW ------------------
    eps_B_L_min = [
        5.42781972291e-29,
        6.69431099159e-28,
        8.03317318991e-27,
    ]
    # ----------------------------------------------------------------

    return eps_B_L_min
```

Appendix C: the numerical script of the reference solution

Listing 4: challenge57.py

```
#!/usr/bin/env python
"""
challenge57.py -- Challenge 57: LIGO with modified (doped) mirrors for ultralight vector dark matter
==========================================================================================

Question.  A vector field A^mu couples to the B-L current, L contains -eps e J^mu_{B-L} A_mu, and is all
    the
dark matter.  LIGO: strain ASD 3e-24 Hz^-1/2 at 250 Hz.  Inner mirrors (ITMs) have Q_D/M = 0.5/m_n;
the doped outer mirrors (ETMs) gain delta(Q_D/M) = delta q/m_n, delta q = {0.074, 6e-3, 5e-4}.
With T = 13 yr and SNR = 1, what is the smallest eps_{B-L} at 250 Hz for each delta q?

Physics chain (equation numbers refer to solution.tex)
    Step 1  field: omega -> m_A; dark electric field amplitude sqrt(2 rho/eps0) (SI)
    Step 2  force on a mirror: a = eps e (Q/M) sqrt(2 rho/eps0) (e_hat . polarization); free-mass
    displacement dx = a/omega^2
    Step 3  exact round trip of one arm with different ITM/ETM charge-to-mass ratios (Morisaki+21 Eq. 11,
    Michimura+20 Eq. 9):
            dL(t) = [4 sin^2(omega L/2c) + 2 delta q/q0] n.dx(t-L/c)  [in phase]  + 2 L (n.k)(n.dx)  [
    quadrature]
            = undoped finite-light-travel-time term + doping term + k.dr term
    Step 4  Michelson output h = -(dL_n - dL_m)/(2L); RMS over random polarization and propagation
    direction
            (<(n.e)^2> = 1/3, n.m = 0):  <h^2> = (dx0^2 / 4L^2) [Gamma^2/3 + (2 k L)^2/9]
            (reduces to Morisaki+21 Eqs. 23-24 for delta q = 0: checked under --verbose)
    Step 5  detection statistics: power threshold <h^2> = S_h / T_eff, T_eff = sqrt(tau T), tau = 2 pi/(
    m_A v^2)
            (Pierce+18 Eq. 9, Morisaki+21 Eqs. 20-21: the convention of the problem's source literature);
            alternative: amplitude matched-filter threshold with the Derevianko R(T) as in Challenge 56
    Step 6  eps_min = sqrt(S_h/(T_eff <h^2>_1)),  <h^2>_1 = <h^2> per unit eps^2

Reproducibility: PROBLEM (verbatim inputs) / MODEL (non-given inputs with sources) / CONV; astropy CODATA
    2018;
--verbose prints intermediate quantities and checks; results.json and figs/ written in the working
    directory.
Run:  conda run -n pyjax python challenge57.py [--verbose] [--no-fig]
"""
import sys, argparse, json
import numpy as np
import astropy
import astropy.units as u
import astropy.constants as const
from scipy.integrate import quad

# ==========================================================================
# INPUTS
# ==========================================================================
PROBLEM = dict(                                         # verbatim from problem.tex
    f_DM     = 250.0 * u.Hz,
    sqrt_Sh  = 3.0e-24 / u.Hz**0.5,                     # LIGO strain ASD at 250 Hz (one-sided)
    q0       = 0.5,                                     # inner mirrors: Q_D/M = q0/m_n
    delta_q  = [0.074, 6e-3, 5e-4],                     # outer mirrors gain delta q/m_n
    T_obs    = 13.0 * u.yr,                             # Julian years (astropy)
    SNR      = 1.0,
)
MODEL = dict(                                           # NOT given by the problem
    rho_DM   = 0.4 * u.GeV / u.cm**3,   # Pierce+18, Morisaki+21, LVK 2022 (Read 2014); carried over from
    Challenge 56
```

```
    v_DM     = 230.0 * u.km / u.s,      # carried over from Challenge 56; Pierce+18 / Morisaki+21 /
    Michimura+20 (RAVE, Smith+07)
    L_arm    = 3994.5 * u.m,            # Advanced LIGO arm length (Aasi+15, CQG 32, 074001, Table 1); "4
    km" variant below
    n_dot_m  = 0.0,                     # perpendicular arms
    rho_alt  = 0.3 * u.GeV / u.cm**3,   # direct-detection convention (Lewin & Smith; Baxter+21) ->
    variant
    L_alt    = 4000.0 * u.m,            # nominal "4 km" -> variant
)
CONV = dict(
    threshold = "power: <h^2> = S_h/T_eff, T_eff = sqrt(tau T), tau = 2 pi/(m_A v^2) (Pierce+18 Eq. 9;
    Morisaki+21 Eqs. 20-21)",
    averaging = "RMS over random polarization and propagation direction (all source papers); forced by 13-
    yr sidereal averaging",
    alt_stat  = "amplitude matched filter h0 sqrt(T/S_h) = 1 with Derevianko R(T), xi c = v, eta = 1 (as
    in Challenge 56)",
)

hbar, c, eps0, e_ch, m_n = const.hbar, const.c, const.eps0, const.e.si, const.m_n


def banner(s): print("\n" + "=" * 78 + "\n" + s + "\n" + "=" * 78)


# ============================================================================
def step1_field(P, M, verbose):
    omega = (2 * np.pi * P["f_DM"]).to(1 / u.s)
    mA = (hbar * omega).to(u.eV)                                   # m_A c^2
    rho = M["rho_DM"].to(u.J / u.m**3)                               # GeV/cm^3 -> J/m^3 (energy density)
    E_D = np.sqrt(2 * rho / eps0).to(u.V / u.m)                     # dark electric field amplitude
    beta = (M["v_DM"] / c).decompose().value
    k = (omega / c * beta).to(1 / u.m)                              # m_A v / hbar
    if verbose:
        banner("Step 1: field")
        print(f"omega = {omega:.5e}; m_A c^2 = {mA:.5e}; rho = {rho:.5e}; E_D = sqrt(2 rho/eps0) = {E_D:.5
    e}")
        print(f"beta = {beta:.5e}; k = {k:.4e}; k L = {(k*M['L_arm']).decompose():.4e}")
    return dict(omega=omega, mA=mA, rho=rho, E_D=E_D, beta=beta, k=k)


# ============================================================================
def step2_displacement(P, M, F, verbose):
    """Displacement amplitude of an inner mirror per unit eps: dx0 = e (q0/m_n) E_D / omega^2."""
    a0 = (e_ch * P["q0"] / m_n * F["E_D"]).to(u.m / u.s**2)          # per unit eps
    dx0 = (a0 / F["omega"]**2).to(u.m)
    if verbose:
        banner("Step 2: force and displacement per unit eps")
        print(f"e q0/m_n = {(e_ch*P['q0']/m_n).to(u.C/u.kg):.5e}; a0 = {a0:.5e}; dx0 = {dx0:.5e}")
    return dict(a0=a0, dx0=dx0)


# ============================================================================
def step34_signal(P, M, F, D, verbose, L=None, include_ltt=True, include_k=True):
    """<h^2> per unit eps^2 for each delta q.  Exact round trip, RMS over polarization and direction."""
    L = M["L_arm"] if L is None else L
    x = (F["omega"] * L / c).decompose().value                      # omega L / c
    ltt = 4 * np.sin(x / 2)**2 if include_ltt else 0.0              # undoped finite-light-travel-time
    coefficient
    kL = (F["k"] * L).decompose().value
    dx0 = D["dx0"].to(u.m).value; Lm = L.to(u.m).value
    out = {}
    for dq in P["delta_q"]:
```

```
        Gam = ltt + 2 * dq / P["q0"]                               # in-phase coefficient Gamma (round
    trip)
        quad_term = (2 * kL)**2 / 9 if include_k else 0.0
        h2 = dx0**2 / (4 * Lm**2) * (Gam**2 / 3 + quad_term)          # <h^2> per eps^2 (time +
    orientation average)
        out[dq] = dict(Gamma=Gam, h2=h2, ltt=ltt, dop=2 * dq / P["q0"], kL=kL)
    if verbose:
        banner("Steps 3-4: arm signal and Michelson output per unit eps")
        print(f"omega L/c = {x:.5e}; 4 sin^2(omega L/2c) = {ltt:.4e}; 2 k L = {2*kL:.4e}")
        for dq, r in out.items():
            print(f"delta q = {dq:.3g}: doping coeff 2dq/q0 = {r['dop']:.4e}, Gamma = {r['Gamma']:.4e}, "
                  f"undoped/doping = {r['ltt']/r['dop']:.4f}, <h^2>^(1/2) per eps = {np.sqrt(r['h2']):.4e}
    ")
        # check against Morisaki Eqs. (23)-(24) in the undoped limit.  Their eps^2 e^2 rho (Q/M)^2 (
    natural units)
        # is a0^2/2 in SI (a0 = e (Q/M) sqrt(2 rho/eps0)), and their m_A -> omega, k = m_A v -> k here.
        a0 = D["a0"].value; om = F["omega"].value
        mor23 = 8 * (a0**2 / 2) * np.sin(x / 2)**4 / (3 * om**4 * Lm**2) * (1 - M["n_dot_m"])
        mor24 = 2 * (a0**2 / 2) * (F["beta"] / (om * c.value))**2 / 9 * (1 - M["n_dot_m"]**2)   # 2 eps^2
    e^2 v^2 rho (Q/M)^2/(9 m^2), SI: /c^2
        mine23 = dx0**2 / (4 * Lm**2) * (ltt**2 / 3); mine24 = dx0**2 / (4 * Lm**2) * (2 * kL)**2 / 9
        print(f"check vs Morisaki Eq.(23): mine {mine23:.6e} vs {mor23:.6e} (ratio {mine23/mor23:.6f}); "
              f"Eq.(24): mine {mine24:.6e} vs {mor24:.6e} (ratio {mine24/mor24:.6f})")
    return out


# ============================================================================
def A2(tau, tau_c, eta):
    x2 = (tau / tau_c)**2
    return np.exp(-eta**2 * x2 / (1 + x2)) / (1 + x2)**1.5


def R_of_T(T, tau_c, eta=1.0):
    I, _ = quad(lambda tt: (1 - tt / T)**2 * A2(tt, tau_c, eta), 0, T, limit=500,
                points=[tau_c, 3 * tau_c, 10 * tau_c] if T > 10 * tau_c else None)
    return ((T / 3) / I)**0.25


def step56_threshold(P, M, F, verbose):
    Sh = (P["sqrt_Sh"]**2).to(1 / u.Hz).value
    T = P["T_obs"].to(u.s).value
    tau = 1 / (P["f_DM"].value * F["beta"]**2)                          # 2 pi/(m v^2)
    tau_c = tau / (2 * np.pi)                                            # 1/(xi^2 omega)
    Teff = np.sqrt(tau * T)
    h2_thr = Sh / Teff                                                   # power threshold on <h^2>
    R = R_of_T(T, tau_c, 1.0)
    h0_thr_alt = np.sqrt(Sh / T) * R                                     # amplitude threshold (56-style)
    if verbose:
        banner("Steps 5-6: thresholds")
        print(f"T = {T:.4e} s; tau = 2pi/(m v^2) = {tau:.1f} s; tau_c = {tau_c:.1f} s; T/tau = {T/tau:.3e
    }; T_eff = {Teff:.4e} s")
        print(f"power threshold <h^2> = S/T_eff = {h2_thr:.4e} (sqrt = {np.sqrt(h2_thr):.4e})")
        print(f"amplitude alternative: R(T) = {R:.4f}; h0 = sqrt(S/T) R = {h0_thr_alt:.4e}  (<h^2> = h0
    ^2/2 = {h0_thr_alt**2/2:.4e})")
    return dict(Sh=Sh, T=T, tau=tau, tau_c=tau_c, Teff=Teff, h2_thr=h2_thr, R=R, h0_thr_alt=h0_thr_alt)


# ============================================================================
def main():
    ap = argparse.ArgumentParser(); ap.add_argument("--verbose", action="store_true"); ap.add_argument("--
    no-fig", action="store_true")
    a = ap.parse_args(); V = a.verbose
```

```
    if V:
        banner("Environment"); print(f"astropy {astropy.__version__} ({const.hbar.reference})")
        print("PROBLEM:", {k: str(v) for k, v in PROBLEM.items()}); print("MODEL:", {k: str(v) for k, v in
     MODEL.items()}); print("CONV:", CONV)
    F = step1_field(PROBLEM, MODEL, V)
    D = step2_displacement(PROBLEM, MODEL, F, V)
    S = step34_signal(PROBLEM, MODEL, F, D, V)
    Th = step56_threshold(PROBLEM, MODEL, F, V)

    # variants of the signal model
    S_dop = step34_signal(PROBLEM, MODEL, F, D, False, include_ltt=False, include_k=False)   # doping only
     (intended)
    S_nok = step34_signal(PROBLEM, MODEL, F, D, False, include_k=False)
    S_L4 = step34_signal(PROBLEM, MODEL, F, D, False, L=MODEL["L_alt"])
    rho_ratio = (MODEL["rho_alt"] / MODEL["rho_DM"]).decompose().value

    banner("RESULT: smallest eps_{B-L} at SNR = 1, f = 250 Hz, T = 13 yr")
    rows = []
    hdr = f"{'signal model':46s} {'statistic':30s} " + " ".join(f"{'dq='+str(dq):>12s}" for dq in PROBLEM[
    "delta_q"])
    print(hdr); print("-" * len(hdr))
    def eps_power(Sig, h2thr=Th["h2_thr"]): return [np.sqrt(h2thr / Sig[dq]["h2"]) for dq in PROBLEM["
    delta_q"]]
    def eps_amp(Sig): return [Th["h0_thr_alt"] / np.sqrt(2 * Sig[dq]["h2"]) for dq in PROBLEM["delta_q"]]
    table = [
        ("exact (ltt + doping + k.dr), L = 3994.5 m", "power, T_eff = sqrt(tau T)", eps_power(S)),
        ("exact", "amplitude MF + Derevianko R(T)", eps_amp(S)),
        ("doping only (as the problem implies)", "power, T_eff = sqrt(tau T)", eps_power(S_dop)),
        ("exact, no k.dr term", "power", eps_power(S_nok)),
        ("exact, L = 4000 m", "power", eps_power(S_L4)),
        ("exact, rho = 0.3 GeV/cm^3", "power", [x / np.sqrt(rho_ratio) for x in eps_power(S)]),
    ]
    for sn, st, vals in table:
        rows.append(dict(signal=sn, statistic=st, eps=vals))
        print(f"{sn:46s} {st:30s} " + " ".join(f"{v:12.4e}" for v in vals))
    json.dump(dict(rows=rows, thresholds={k: float(v) for k, v in Th.items()},
                   signal={str(dq): {k: float(v) for k, v in r.items()} for dq, r in S.items()},
                   dx0=D["dx0"].value, mA_eV=F["mA"].value, kL=float((F["k"]*MODEL["L_arm"]).decompose().
    value)),
              open("results.json", "w"), indent=1)
    print("wrote results.json")

    if not a.no_fig:
        import matplotlib; matplotlib.use("Agg"); import matplotlib.pyplot as plt, os
        os.makedirs("figs", exist_ok=True)
        dqs = np.logspace(-4, -0.5, 200)
        x = (F["omega"] * MODEL["L_arm"] / c).decompose().value; ltt = 4 * np.sin(x / 2)**2
        fig, ax = plt.subplots(figsize=(6, 3.6))
        ax.loglog(dqs, ltt / (2 * dqs / PROBLEM["q0"]), label=r"undoped light-travel-time term / doping
    term")
        for dq in PROBLEM["delta_q"]:
            ax.axvline(dq, color="grey", lw=0.8)
        ax.axhline(1, color="k", ls="--", lw=0.8)
        ax.set_xlabel(r"$\delta q$"); ax.set_ylabel("amplitude ratio"); ax.legend(fontsize=8)
        ax.tick_params(which="major", length=5.25); ax.tick_params(which="minor", length=3)
        fig.tight_layout(); fig.savefig("figs/ltt_vs_doping.png", dpi=160); print("figure: figs/
    ltt_vs_doping.png")


if __name__ == "__main__":
    main()
```